
Predicting Volatility of Daily Bitcoin Returns with ARCH, GARCH, and EGARCH Models

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Abstract

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Keywords: Bitcoin; volatility forecasting; ARCH; GARCH; EGARCH; heavy-tailed innovations; financial econometrics.

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1 Introduction

Financial markets are characterised by uncertainty. For investors, risk managers and policy makers, this uncertainty is reflected not only in the direction of price changes, but also in the magnitude of those changes. Volatility is therefore one of the central concepts in financial econometrics. It measures the variability of returns over time and is commonly interpreted as an indicator of financial risk. Accurate volatility forecasts are important for portfolio allocation, risk management, derivative pricing and trading decisions, since they provide information about the expected degree of future market uncertainty.

Bitcoin provides a particularly relevant setting for volatility modelling. Since its introduction in 2009, Bitcoin has developed from a relatively small digital asset into a widely traded cryptocurrency and investment instrument. In contrast to traditional financial assets, Bitcoin is not issued by a central authority, is traded continuously and is strongly influenced by speculative demand, liquidity conditions, investor sentiment and regulatory developments. These characteristics make Bitcoin returns highly volatile and motivate the use of econometric models that can describe and forecast changes in conditional volatility over time.

A key empirical regularity of financial return series is that volatility is time-varying. In particular, returns often exhibit volatility clustering, meaning that periods with large absolute returns tend to be followed by further periods with large absolute returns, while periods with small absolute returns tend to be followed by further periods with small absolute returns (Mandelbrot, 1963; Cont, 2001). This implies that, although the sign of future returns may be difficult to predict, the magnitude of future returns may still be partly predictable from past information. A model with constant variance would treat calm and turbulent market periods as equally risky, which is unlikely to be appropriate for Bitcoin returns. Consequently, the conditional variance of returns should be modelled as a time-varying process.

The Autoregressive Conditional Heteroskedasticity model introduced by Engle (1982) provides a natural starting point for modelling time-varying volatility. In an ARCH model, the conditional variance depends on past squared shocks. Hence, large shocks in previous periods increase the predicted variance in the current period. This feature allows the model to capture volatility clustering. However, a pure ARCH model may require many lags to describe the persistence typically observed in financial volatility. The Generalized Autoregressive Conditional Heteroskedasticity model of Bollerslev (1986) extends the ARCH framework by allowing the conditional variance to depend not only on past squared shocks, but also on lagged conditional variances. This makes the GARCH model a parsimonious and widely used model for persistent volatility dynamics.

Although ARCH and GARCH models capture volatility clustering, standard specifications are symmetric: positive and negative shocks of the same magnitude have the same effect on future volatility. This may be restrictive, since financial markets can react differently to negative and positive news. In equity markets, negative shocks are often associated with stronger increases in future volatility, a phenomenon commonly referred to as the leverage effect. Whether a similar asymmetric response is present in Bitcoin returns is an empirical question. The Exponential GARCH model proposed by Nelson (1991) is useful in this context because it models the logarithm of the conditional variance and allows positive and negative shocks to have different effects on future volatility.

This thesis investigates the conditional volatility of daily Bitcoin log returns using ARCH, GARCH and EGARCH models across different innovation distributions. The main research question is:

Which ARCH-family specification, combined with an appropriate innovation distribution, provides the most accurate description and forecast of the conditional volatility of daily Bitcoin returns?

To answer the research question, this thesis proceeds as follows. Section 2 reviews the relevant literature on financial return volatility, ARCH-family models, conditional innovation distributions and cryptocurrency volatility. Section 3 describes the Bitcoin price data, explains the construction of daily log returns and examines the main empirical properties of the return series, including stationarity, non-normality, volatility clustering and ARCH effects. Section 4 introduces the empirical **modelling** framework, specifies the ARCH, GARCH and EGARCH models, discusses the conditional innovation distributions and explains the estimation, diagnostic and model comparison procedures. Section 5 presents the in-sample estimation results, including the information-criterion ranking, parameter estimates of the selected in-sample model and residual diagnostics. Section 6 evaluates the out-of-sample volatility forecasts and compares the forecasting performance of the estimated models. Section 7 discusses the main findings in relation to the literature and reflects on the limitations of the empirical approach. Section 8 concludes by answering the research question, while Section 9 outlines directions for future research.

2 Literature Review

2.1 Stylized facts of financial returns

Financial return series typically display empirical regularities that distinguish them from independent and identically distributed Gaussian observations. These regularities are often referred to as stylized facts: statistical properties that are not exact laws, but that appear consistently across many financial markets and sample periods (Cont, 2001). Since this thesis studies the volatility of Bitcoin returns, these stylized facts provide the empirical motivation for using conditional volatility models rather than models with a constant unconditional variance.

A first important stylized fact is that financial prices themselves are often non-stationary, whereas return series are usually more suitable for time-series **modelling**. For this reason, empirical studies typically analyse returns rather than price levels. In the context of this thesis, Bitcoin prices are transformed into log returns, so that the object of interest is the relative price change from one day to the next. This transformation removes the strong level dependence present in the price series and produces a series that can be **analysed** using standard tools for stationary financial time series. Whether the resulting Bitcoin return series is indeed stationary is assessed empirically in the data section.

A second stylized fact is that financial returns are usually not normally distributed. In particular, return distributions tend to exhibit excess kurtosis, meaning that extreme positive and negative observations occur more frequently than would be expected under a Gaussian distribution. This property is commonly referred to as fat-tailedness or leptokurtosis. It is important for volatility **modelling** because a model with conditionally normal innovations may underestimate the probability of large return movements. Consequently, empirical volatility studies often compare Gaussian innovations with heavier-tailed alternatives, such as the Student- t distribution.

A third stylized fact is that raw returns often display little linear autocorrelation, especially at daily frequencies. This means that past returns usually contain limited information about the sign or direction of future returns. This observation is consistent with the idea that financial markets quickly incorporate new information into prices. However,

the absence of strong autocorrelation in returns does not imply that returns are independent. In particular, nonlinear transformations of returns, such as absolute returns or squared returns, often show much stronger serial dependence (Cont, 2001). This distinction is central for volatility **modelling**: the conditional mean of returns may be difficult to predict, while the conditional variance may still be predictable.

A fourth and particularly important stylized fact is volatility clustering. This refers to the tendency of large price changes to be followed by further large price changes, and small price changes to be followed by further small price changes. Early empirical work by Mandelbrot (1963) and Fama (1965) already emphasized that large changes in speculative prices tend to cluster together, although the direction of those changes remains difficult to predict. In volatility terms, this means that periods of high uncertainty tend to persist for some time, while calm periods also tend to persist. Volatility clustering therefore suggests that the conditional variance of returns changes over time and depends on past information.

A final stylized fact relevant for this thesis is the possible asymmetry in volatility responses. In equity markets, negative returns are often found to have a stronger effect on future volatility than positive returns of the same magnitude. This is frequently referred to as the leverage effect. For Bitcoin, however, the traditional leverage interpretation is less direct, because Bitcoin is not a firm with a debt-to-equity structure. Nevertheless, it remains possible that negative and positive Bitcoin return shocks have different effects on subsequent volatility. For this reason, it is more precise in this thesis to refer to asymmetric volatility responses or asymmetric news effects rather than to interpret such asymmetry mechanically as financial leverage.

Together, these stylized facts motivate the use of ARCH-family models (Bollerslev et al., 1992). The following subsections therefore discuss these models and their relevance for **modelling** and forecasting Bitcoin return volatility.

2.2 ARCH models and conditional heteroskedasticity

The stylized facts discussed above suggest that financial return volatility is not constant over time. In particular, volatility clustering implies that large shocks tend to be followed by periods of high volatility, while small shocks tend to be followed by periods of relatively low volatility. This observation is difficult to reconcile with classical time-series models in which the one-period-ahead forecast variance is assumed to be constant.

Engle (1982) introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model to allow the conditional variance of a time series to depend on past information. The central idea is that recent shocks contain information about current risk. If a large return shock occurred in the previous period, the model predicts a higher conditional variance in the current period. In this way, the ARCH framework provides a formal econometric explanation for volatility clustering.

A useful feature of ARCH models is the distinction between conditional and unconditional variance. The unconditional variance may be constant under suitable stationarity conditions, while the conditional variance is allowed to change over time as new information arrives. This distinction is particularly important for financial returns, since raw returns often exhibit little linear autocorrelation, whereas squared or absolute returns tend to show substantial serial dependence. ARCH models are therefore designed not primarily to predict the sign of future returns, but to model the changing magnitude of return fluctuations.

The ARCH framework also provides the basis for testing whether conditional heteroskedasticity is present. Engle’s ARCH–LM test examines whether squared residuals

are serially dependent. If the null hypothesis of no ARCH effects is rejected, this indicates that past shocks help explain current volatility and that a constant-variance model is inadequate. For this reason, the ARCH–LM test is used in many empirical volatility studies as a preliminary diagnostic before estimating ARCH-family models.

Despite its importance, a pure ARCH specification has a practical limitation. Financial volatility is often highly persistent, meaning that the effect of a shock can last for many periods. Capturing this persistence with an ARCH model may require many lagged squared shocks, which increases the number of parameters and reduces parsimony. In this thesis, ARCH models are therefore included as simple benchmark specifications. They capture the basic mechanism of volatility clustering, but they are expected to be less parsimonious than GARCH models when volatility persistence is strong.

2.3 GARCH models and volatility persistence

Bollerslev (1986) extended the ARCH framework by introducing the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model. The main innovation of the GARCH model is that current conditional volatility depends not only on past squared shocks, as in an ARCH model, but also on past conditional volatility itself. This makes the model more flexible and more parsimonious when volatility is persistent.

The intuition behind GARCH is that financial markets often remain in high- or low-volatility states for several consecutive periods. A large return shock increases volatility directly, but high volatility can also persist because yesterday’s predicted volatility carries over into today’s volatility. The model therefore separates two related effects: the short-run impact of new shocks and the persistence of past volatility. In empirical financial applications, the persistence component is often substantial, which explains why relatively simple GARCH specifications are widely used for daily return data.

The GARCH(1,1) model has become a standard benchmark in financial econometrics. Although it is a low-order model, it is often able to capture the slowly decaying dependence in squared returns that would otherwise require a high-order ARCH model. This parsimonious representation is one reason why GARCH models are commonly used in risk management, portfolio allocation and volatility forecasting.

For this thesis, the GARCH framework is relevant because Bitcoin returns display clear evidence of volatility clustering and persistent dependence in squared returns. A GARCH model allows this persistence to be captured with relatively few parameters. Comparing GARCH with simpler ARCH specifications therefore makes it possible to examine whether explicitly modelling volatility persistence improves the description and forecasting of Bitcoin return volatility.

2.4 EGARCH models and asymmetric volatility responses

Standard ARCH and GARCH models are symmetric in the sense that positive and negative return shocks of the same magnitude have the same effect on future volatility. This restriction may be too strong for financial return series. Empirical studies often find that bad news and good news affect volatility differently, with negative shocks frequently leading to stronger increases in future volatility than positive shocks of comparable size.

Nelson (1991) introduced the Exponential GARCH (EGARCH) model to account for such asymmetric volatility responses. The EGARCH framework allows the effect of a shock on future volatility to depend on both its size and its sign. This makes it possible to test whether negative and positive return innovations have different implications for subsequent risk. In equity markets, this type of asymmetry is often linked to the leverage

effect, where a fall in equity value increases the debt-to-equity ratio of a firm and thereby raises its riskiness.

For Bitcoin, the traditional leverage interpretation should be used with caution because Bitcoin is not a firm with a debt-to-equity structure. Nevertheless, asymmetric volatility responses may still arise if market participants react differently to **favourable** and **unfavourable** news, regulatory announcements, liquidity shocks or changes in investor sentiment. It is therefore more precise in this thesis to interpret asymmetry as an asymmetric news effect rather than as direct evidence of financial leverage.

A further advantage of EGARCH is that it models volatility in logarithmic form. This automatically ensures that the conditional variance remains positive, without imposing the same non-negativity restrictions that are required in standard ARCH and GARCH models. This feature makes EGARCH a flexible alternative when both persistence and asymmetry may be present.

2.5 Conditional innovation distributions

ARCH-family models describe the evolution of conditional volatility over time, but they do not by themselves determine the full conditional distribution of returns. An additional **modelling** choice concerns the distribution of the standardized innovations. While the volatility model determines the **behaviour** of the conditional volatility, the innovation distribution determines the probabilistic **behaviour** of the shocks driving returns.

Early ARCH-family models were often estimated under the assumption of conditionally normally distributed innovations. For example, the original ARCH model of Engle (1982) was developed using a Gaussian likelihood framework. However, financial return series often exhibit excess kurtosis and occasionally skewness, implying that extreme observations occur more frequently than predicted by a Gaussian distribution (Cont, 2001). Subsequent work therefore considered heavier-tailed alternatives, such as the Student- t distribution, for speculative returns (Bollerslev, 1987).

A widely used alternative is the Student- t distribution, which allows for heavier tails than the normal distribution and is therefore better able to capture large return movements. More flexible specifications additionally allow for asymmetry in the innovation distribution. Examples include the skewed Student- t distribution and other asymmetric heavy-tailed distributions. Such distributions may improve model fit when return series exhibit both fat tails and skewness.

The Generalized Error Distribution (GED), also known as the exponential power distribution, provides a flexible symmetric alternative to the normal distribution (Zhu and Zinde-Walsh, 2009). Its shape parameter controls the peakedness and tail thickness of the distribution, allowing it to represent both heavier- and thinner-tailed **behaviour** relative to the Gaussian case. In particular, the GED nests the normal distribution for a specific value of the shape parameter and the Laplace distribution for another. This flexibility makes the GED relevant for financial return series, where excess kurtosis is frequently observed.

For this thesis, the innovation distribution constitutes an additional dimension of model comparison alongside the volatility specification itself. The empirical analysis therefore considers the normal, Student- t distribution, GED, and skewed Student- t distributions as alternative specifications for the standardized innovations.

2.6 Volatility **modelling** of Bitcoin and cryptocurrencies

The ARCH-family models discussed above were originally developed for traditional economic and financial time series, but they have also been widely applied to cryptocurrencies. This is natural because cryptocurrency returns often display the same empirical features that motivate conditional volatility models, such as large price movements, volatility clustering and heavy-tailed return distributions. As the first and most widely studied cryptocurrency, Bitcoin has received particular attention in the volatility **modelling** literature.

One of the earlier applications is Dyhrberg (2016), who studies Bitcoin using GARCH-type models and compares its volatility **behaviour** with that of gold and the US dollar. The study argues that Bitcoin shares some characteristics with both assets and may have relevance for risk management. This paper is important because it treats Bitcoin not merely as a technological innovation, but as a financial asset whose risk dynamics can be studied using standard tools from financial econometrics.

A more direct comparison of volatility models for Bitcoin is provided by Katsiampa (2017). This study compares several GARCH-type models and finds that an autoregressive component GARCH model provides the best fit according to goodness-of-fit criteria. The result suggests that Bitcoin volatility may contain both short-run and long-run components. However, the model set considered by Katsiampa (2017) is broader than the model set used in this thesis. The present thesis focuses specifically on ARCH, GARCH and EGARCH models in order to provide a more transparent comparison between a basic conditional heteroskedasticity model, a persistent volatility model and an asymmetric volatility model.

The cryptocurrency volatility literature was extended by Chu et al. (2017), who apply GARCH-type models to seven major cryptocurrencies. Their study shows that conditional volatility **modelling** is not only relevant for Bitcoin, but also for cryptocurrency markets more generally. By estimating multiple GARCH specifications and assessing model fit and Value-at-Risk performance, they demonstrate that the choice of both volatility model and innovation distribution can materially affect empirical conclusions.

The paper most closely related to this thesis is Yildirim and Bekun (2023), who compare ARCH, GARCH and EGARCH models for Bitcoin returns. They use weekly Bitcoin price data, compute logarithmic returns, test for stationarity using the augmented Dickey–Fuller test and examine ARCH effects before estimating conditional variance models. Their findings indicate that the GARCH(1,1) model performs best according to the Akaike information criterion. This provides a useful benchmark for the present thesis. However, their use of weekly data differs from the daily return frequency considered here, and volatility dynamics may depend on the sampling frequency. Therefore, their results do not remove the need for a separate empirical analysis of daily Bitcoin returns.

Overall, the existing literature supports the use of ARCH-family models for Bitcoin and cryptocurrency volatility. At the same time, previous studies differ in their sample periods, data frequencies, model classes, innovation distributions and evaluation criteria. This thesis builds on this literature by comparing ARCH, GARCH and EGARCH models for daily Bitcoin log returns, while also considering alternative innovation distributions and both in-sample and out-of-sample model evaluation.

3 Data

3.1 Bitcoin price data

This thesis uses daily Bitcoin price data obtained from Yahoo Finance through the `yfinance` Python package. An overview of the dataset is provided in Table 1. The sample spans multiple market phases and is therefore well suited for the analysis of Bitcoin volatility.

Table 1: Overview of the Bitcoin dataset

Characteristic	Description
Asset	Bitcoin (BTC-USD)
Data source	Yahoo Finance (<code>yfinance</code>)
Frequency	Daily
Currency	United States Dollar (USD)
Sample start	1 January 2016
Sample end	1 June 2026
Observations	3,804
Price variable	Daily closing price
Return measure	Daily log returns

Daily observations are used because they provide a suitable balance between information content and data quality. They are frequent enough to capture time variation in volatility, while avoiding many of the market microstructure effects that may affect higher-frequency data. Since Bitcoin is traded continuously, the sample consists of daily calendar observations rather than observations restricted to weekday trading days. The price variable is the daily closing price of BTC-USD. The daily closing prices are then transformed into log returns, since the objective of this thesis is to model and forecast return volatility rather than the level of Bitcoin prices. Taking first differences of log prices reduces the sample by one observation, resulting in 3,803 daily log returns. The construction of these returns is described in [section 3.2](#).

Figure 1 presents the daily Bitcoin closing price over the sample period. The series shows substantial changes in the level of Bitcoin prices, including episodes of rapid appreciation and sharp declines.

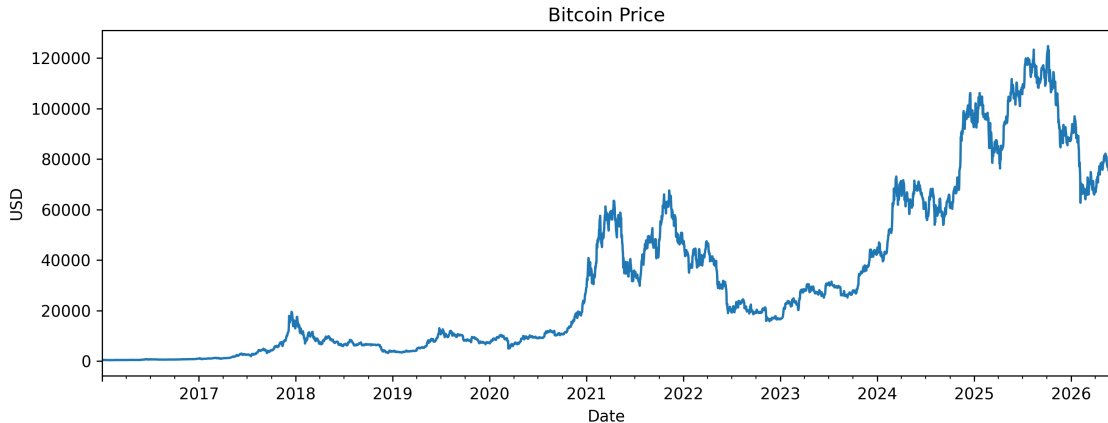


Figure 1: Daily Bitcoin price series from 1 January 2016 to 1 June 2026.

3.2 Construction of log returns

The raw Bitcoin price series is not used directly in the empirical analysis. Instead, prices are transformed into daily log returns, which measure proportional price changes and are the standard variable used in financial volatility **modelling**.

Let p_t denote the Bitcoin closing price observed on day t , where $t = 1, 2, \dots, T$ and T denotes the total number of price observations in the sample. The raw daily log return is defined as

$$r_t = 100 (\log(p_t) - \log(p_{t-1})), \quad (1)$$

for $t = 2, \dots, T$. The multiplication by 100 expresses returns approximately in percentage terms.

Log returns are widely used in the financial econometrics literature for several reasons. First, they measure proportional rather than absolute price changes, making returns comparable across different price levels. Second, continuously compounded returns possess desirable statistical properties and are additive over time. Finally, financial volatility models such as ARCH, GARCH and EGARCH are typically specified in terms of return series rather than price levels (Tsay, 2005).

In the empirical volatility models, the raw return r_t is decomposed into a constant mean component and a mean-adjusted return. This will be introduced more formally in **Section 4.1**.

The transformation from prices to returns substantially alters the appearance of the series. While the Bitcoin price series exhibits pronounced long-run movements and periods of sustained growth or decline, the raw return series fluctuates around a relatively stable level. Figure 2 shows that periods of large positive and negative returns tend to occur close together, while calmer periods are also visible. This provides a first visual indication of volatility clustering, a feature that is examined more formally in **Section 3.5**.

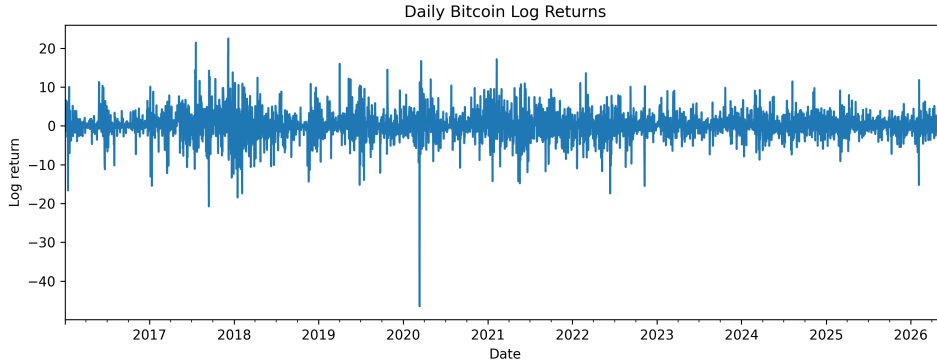


Figure 2: Daily Bitcoin log returns from 1 January 2016 to 1 June 2026.

3.3 Descriptive statistics

Table 2 reports summary statistics for the raw daily Bitcoin log return series defined in (1). These statistics provide a first indication of the unconditional distributional properties of the returns before the time-series dependence and conditional volatility dynamics are **analysed**.

Table 2: Descriptive statistics and Jarque–Bera test of raw daily Bitcoin log returns

Statistic	Value
Observations	3,803
Mean	0.1350
Standard deviation	3.5029
Minimum	-46.4730
Maximum	22.5119
Skewness	-0.6636
Pearson kurtosis	15.3239
Jarque–Bera statistic	24276.12
JB p -value	< 0.0001

The average raw daily log return is approximately 0.135, indicating a positive mean return over the sample period. However, this average is small relative to the standard deviation of 3.503, which reflects the substantial variability of daily Bitcoin returns. The minimum and maximum values further illustrate the presence of extreme daily return realisations. The most negative raw daily log return equals -46.473, whereas the largest positive raw daily log return equals 22.512. These observations indicate that the return series contains large shocks, which is a common feature of financial return data and is particularly relevant for volatility **modelling**.

The distribution of returns also deviates strongly from normality. The skewness is negative, with a value of -0.664, implying that the left tail of the distribution is more pronounced than the right tail. In addition, the Pearson kurtosis equals 15.324, which is far above the normal-distribution benchmark of three. This indicates that Bitcoin returns are leptokurtic: the distribution is more peaked around its centre and has heavier tails than a normal distribution. Hence, extreme observations occur more frequently than would be expected under normality.

This conclusion is supported by the Jarque–Bera test, which jointly tests whether the skewness and kurtosis of the series are consistent with a normal distribution. The test statistic is 24276.12 and the corresponding p -value is smaller than 0.0001. Therefore, the null hypothesis of normally distributed returns is strongly rejected. This provides formal evidence that the unconditional distribution of Bitcoin returns is non-normal.

Overall, the descriptive statistics indicate that Bitcoin log returns are asymmetric and heavy-tailed. These distributional features are consistent with the stylized facts discussed in **Section 2.1** and motivate the consideration of non-normal innovation distributions, such as the Student- t and skewed Student- t distributions, in the volatility models estimated later.

3.4 Stationarity analysis

Before **modelling** the conditional variance, the time-series properties of the return series are examined. In particular, it is important to verify that the log returns do not contain a unit root, since ARCH-family models are applied to stationary return processes rather than non-stationary price levels.

The Augmented Dickey–Fuller (ADF) test is used for this purpose. The null hypothesis is that the series contains a unit root, while the alternative hypothesis is stationarity (Dickey and Fuller, 1979). Table 3 reports the results of the ADF test applied to the raw daily Bitcoin log return series.

Table 3: Augmented Dickey–Fuller test for raw daily Bitcoin log returns

Statistic	Value
ADF statistic	-42.946
p -value	< 0.001
1% critical value	-3.432
5% critical value	-2.862
10% critical value	-2.567

The lag length is selected automatically by minimizing the Akaike Information Criterion, resulting in one lag. Robustness checks using alternative fixed lag lengths lead to the same conclusion, see Appendix A.2. The ADF statistic is strongly negative and the associated p -value is very small. Consequently, the null hypothesis of a unit root is rejected at all conventional significance levels. This provides strong evidence that the Bitcoin log return series is stationary.

The result is consistent with the discussion in [Section 3.2](#), where the Bitcoin price series was transformed into returns before further analysis. While financial asset prices often exhibit non-stationary [behaviour](#), return series are typically more suitable for econometric [modelling](#). The evidence presented here supports the use of the Bitcoin log return series as the basis for the subsequent volatility analysis.

Having found evidence against a unit root in the return series, the analysis next examines whether returns and squared returns exhibit serial dependence.

3.5 Volatility clustering and ARCH effects

The next step is therefore to examine whether the magnitude of returns displays persistence, which would indicate volatility clustering and conditional heteroskedasticity.

Figure 3 presents the autocorrelation function of the Bitcoin log returns. The autocorrelations are generally small and fluctuate around zero. The Ljung–Box statistics in Table 4 provide similar evidence: at the 5% significance level, the null hypothesis of no autocorrelation is not rejected at the reported lag lengths. This suggests that the return series contains little linear dependence in its conditional mean.

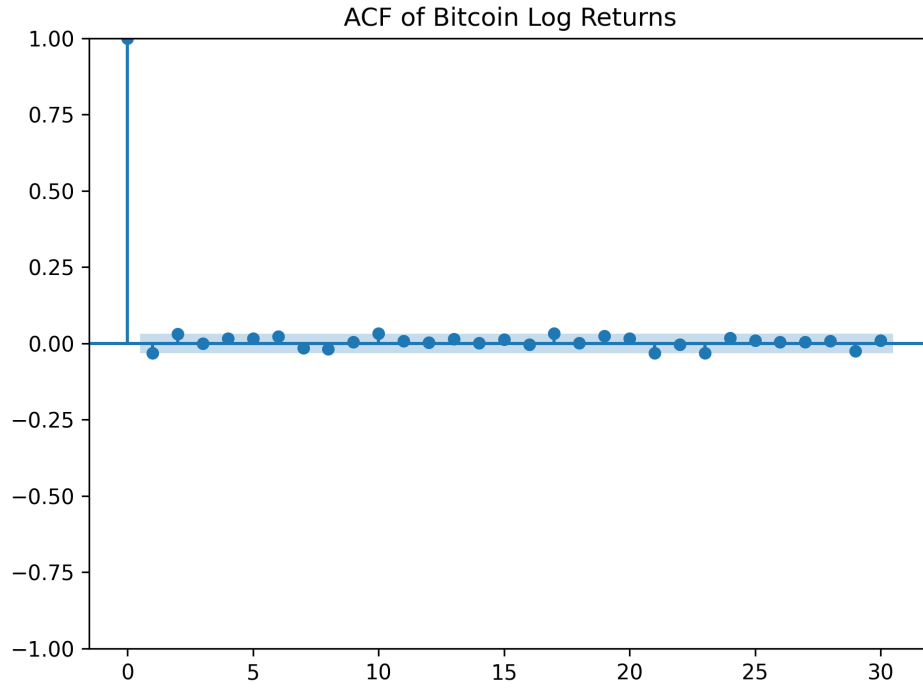


Figure 3: Autocorrelation function of daily Bitcoin log returns.

Table 4: Ljung–Box test for Bitcoin log returns

Lag	Statistic	p-value
10	17.262	0.069
20	26.635	0.146
30	39.116	0.123

The absence of strong autocorrelation in raw returns does not rule out dependence in volatility. To examine this, Figure 4 reports the autocorrelation function of squared Bitcoin log returns. In contrast to the raw returns, the squared returns exhibit positive autocorrelations over several lags, indicating persistence in the magnitude of return shocks.

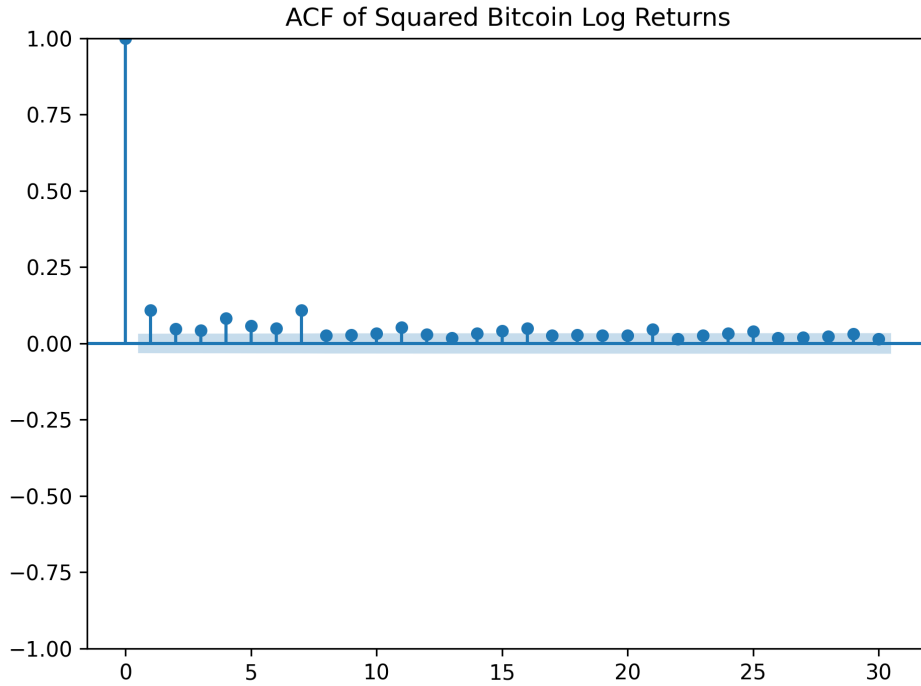


Figure 4: Autocorrelation function of squared daily Bitcoin log returns.

This finding is confirmed by the Ljung–Box test applied to squared returns, reported in Table 5. The null hypothesis of no autocorrelation is strongly rejected at all reported lag lengths. This contrast between weak autocorrelation in returns and strong autocorrelation in squared returns is typical of financial return series and provides evidence of volatility clustering.

Table 5: Ljung–Box test for squared Bitcoin log returns

Lag	Statistic	p-value
10	163.270	6.84×10^{-30}
20	208.889	1.95×10^{-33}
30	240.194	1.17×10^{-34}

The persistence observed in the squared returns is characteristic of volatility clustering, whereby large return movements tend to be followed by further large movements, while relatively calm periods tend to be followed by additional calm periods. This **behaviour** is one of the principal stylized facts of financial returns discussed in **Section 2.1**.

The autocorrelation in squared returns suggests volatility clustering, but the ARCH–LM test provides a direct formal test for conditional heteroskedasticity. The null hypothesis is that no ARCH effects are present, meaning that past squared shocks do not help explain current conditional variance. Table 6 reports the results.

Table 6: ARCH–LM test for Bitcoin log returns using 10 lags

Statistic	Value
LM statistic	118.952
LM p-value	8.25×10^{-21}
F-statistic	12.245
F-test p-value	3.76×10^{-21}

Both the LM statistic and the corresponding F-statistic strongly reject the null hypothesis of no ARCH effects. Hence, the conditional variance of Bitcoin returns cannot be treated as constant over time. Instead, the results indicate that current volatility depends on past return shocks.

Taken together, the evidence from the return plots, squared-return autocorrelations, Ljung–Box tests and ARCH–LM test supports the use of ARCH-family models for Bitcoin return volatility. The next chapter therefore introduces the ARCH, GARCH and EGARCH specifications used to model and forecast the conditional variance.

4 Methodology

4.1 Empirical modelling framework

The empirical analysis is based on the raw daily Bitcoin log returns constructed in Section 3.2. Let p_t denote the daily closing price of Bitcoin on day t , and let r_t denote the corresponding raw daily log return as defined in (1)

To keep the volatility model notation concise, the raw return is decomposed into a constant mean component and a mean-adjusted return:

$$r_t = \mu + y_t,$$

or equivalently,

$$y_t = r_t - \mu.$$

Here, μ denotes the constant mean of the raw return series, while y_t denotes the mean-adjusted return.

Let Y^{t-1} denote the information set generated by past mean-adjusted returns,

$$Y^{t-1} = \{y_{t-1}, y_{t-2}, \dots\}.$$

The observation equation for the mean-adjusted return is defined as

$$y_t = \sigma_t \varepsilon_t, \tag{2}$$

where σ_t denotes the volatility and ε_t is an independent and identically distributed standardized innovation. The innovations satisfy

$$\mathbb{E}(\varepsilon_t) = 0, \quad \text{Var}(\varepsilon_t) = 1,$$

and are assumed to be independent of the past information set Y^{t-1} . It follows that

$$\mathbb{E}(y_t \mid Y^{t-1}) = 0, \quad \text{Var}(y_t \mid Y^{t-1}) = \sigma_t^2.$$

This specification is consistent with the preliminary evidence discussed in [Section 3.5](#). The return series shows little evidence of linear dependence, whereas the squared returns display substantial serial dependence.

Under the conditionally normal benchmark, the observation equation implies

$$y_t \mid Y^{t-1} \sim N(0, \sigma_t^2).$$

When alternative innovation distributions are considered, the conditional variance equation remains unchanged, while the distributional assumption for ε_t is modified.

This framework is common to all volatility models estimated in this thesis. The ARCH, GARCH and EGARCH models differ only in how they specify the evolution of σ_t^2 over time. After estimation, the standardized residuals are obtained as

$$\hat{\varepsilon}_t = \frac{\hat{y}_t}{\hat{\sigma}_t} = \frac{r_t - \hat{\mu}}{\hat{\sigma}_t}, \quad (3)$$

These standardized residuals are used in the diagnostic analysis to assess whether the fitted model has adequately captured the remaining serial dependence in volatility.

4.2 Volatility model specifications

This subsection presents the conditional variance specifications used in the empirical analysis. The models are specified for the mean-adjusted return y_t , defined in [Section 4.1](#). The ARCH, GARCH and EGARCH models differ in how they specify the dynamics of σ_t^2 . All models are estimated under the same mean-adjusted return framework, so that differences in model performance can be attributed to the conditional variance specification and the assumed innovation distribution.

4.2.1 ARCH models

The ARCH model specifies current conditional variance as a function of past squared mean-adjusted returns. In this thesis, ARCH models are included as benchmark specifications because they provide the simplest way to capture volatility clustering. [The](#) ARCH(q) model is given by

$$y_t = \sigma_t \varepsilon_t$$

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i y_{t-i}^2, \quad (4)$$

where $\omega > 0$ and $\alpha_i \geq 0$ for $i = 1, \dots, q$. These restrictions ensure that the conditional variance is strictly positive. Since y_{t-i}^2 measures the size of the return shock in period $t - i$, a positive value of α_i implies that large past shocks increase current volatility.

The ARCH specification allows the conditional variance to vary over time, even though the unconditional variance may be constant under appropriate stationarity conditions. For the ARCH(q) model, a sufficient condition for a finite unconditional variance is

$$\sum_{i=1}^q \alpha_i < 1.$$

Under this condition, volatility shocks are temporary and the variance process reverts to a long-run level. The unconditional variance is then equal to

$$\text{Var}(y_t) = \frac{\omega}{1 - \sum_{i=1}^q \alpha_i}. \quad (5)$$

The empirical analysis estimates ARCH specifications with $q = 1, 2, 3$. These models are denoted by ARCH(1), ARCH(2) and ARCH(3), respectively. The ARCH(1) model is the most parsimonious specification and is given by

$$\sigma_t^2 = \omega + \alpha_1 y_{t-1}^2. \quad (6)$$

The ARCH(2) and ARCH(3) models extend this specification by allowing volatility to depend on additional lagged squared returns:

$$\text{ARCH(2): } \sigma_t^2 = \omega + \alpha_1 y_{t-1}^2 + \alpha_2 y_{t-2}^2, \quad (7)$$

$$\text{ARCH(3): } \sigma_t^2 = \omega + \alpha_1 y_{t-1}^2 + \alpha_2 y_{t-2}^2 + \alpha_3 y_{t-3}^2. \quad (8)$$

Including several ARCH lags allows volatility to respond to shocks over multiple days. However, ARCH models can require many lagged squared returns to capture persistent volatility. This limitation motivates the GARCH model, which introduces lagged conditional variances and is discussed in the next subsection.

4.2.2 GARCH models

The generalized autoregressive conditional heteroskedasticity model, introduced by Bollerslev (1986), extends the ARCH framework by including lagged conditional variances in the volatility equation. This provides a more parsimonious way to model persistent volatility.

A GARCH model with q ARCH terms and p lagged conditional variance terms is specified as

$$y_t = \sigma_t \varepsilon_t$$

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i y_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2, \quad (9)$$

where y_t denotes the mean-adjusted return. The parameters satisfy $\omega > 0$, $\alpha_i \geq 0$ for $i = 1, \dots, q$, and $\beta_j \geq 0$ for $j = 1, \dots, p$. These restrictions ensure that the conditional variance remains positive.

The empirical analysis focuses on the GARCH(1,1) model, which is the standard benchmark specification in applied financial econometrics. It is given by

$$\sigma_t^2 = \omega + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \quad (10)$$

In this model, α_1 measures the short-run effect of new return shocks on volatility, while β_1 measures the persistence of past conditional volatility. The sum $\alpha_1 + \beta_1$ is therefore an important measure of volatility persistence. When this sum is close to one, shocks to volatility decay slowly and high-volatility periods tend to persist.

For the GARCH(1,1) model, a sufficient condition for covariance stationarity is

$$\alpha_1 + \beta_1 < 1.$$

Under this condition, the unconditional variance is finite and equal to

$$\text{Var}(y_t) = \frac{\omega}{1 - (\alpha_1 + \beta_1)}. \quad (11)$$

The parameter ω determines the long-run level of volatility, while α_1 and β_1 determine how quickly the conditional variance responds to and recovers from shocks.

The GARCH(1,1) model is particularly useful in this thesis because Bitcoin returns display volatility clustering and persistence in squared returns. Compared with low-order

ARCH models, GARCH(1,1) can capture longer-lasting volatility dynamics with fewer parameters by allowing past conditional variance to directly influence current conditional variance. The persistence can also be seen by rewriting the GARCH(1,1) model as an infinite-order ARCH process. The full derivation is provided in Appendix A.1. However, the standard GARCH model is symmetric: positive and negative shocks of the same magnitude have the same effect on future volatility because the shock enters the variance equation through y_{t-1}^2 . To allow for asymmetric volatility responses, the next subsection introduces the EGARCH model.

4.2.3 EGARCH models

To allow for asymmetric volatility responses, this thesis also considers the exponential GARCH model introduced by Nelson (1991). While the ARCH and GARCH specifications considered above are symmetric in the sense that positive and negative shocks of the same magnitude have the same effect on future volatility, the EGARCH model allows the sign of a shock to affect subsequent volatility.

Using (2), we can write the standardized innovations as

$$\varepsilon_t = \frac{y_t}{\sigma_t}. \quad (12)$$

In general theoretical notation, the EGARCH(1,1) model is specified as

$$\log(\sigma_t^2) = \omega + \alpha_1 (|\varepsilon_{t-1}| - \mathbb{E}|\varepsilon_{t-1}|) + \gamma_1 \varepsilon_{t-1} + \beta_1 \log(\sigma_{t-1}^2). \quad (13)$$

The term $|\varepsilon_{t-1}| - \mathbb{E}|\varepsilon_{t-1}|$ is centred around zero and captures the effect of the magnitude of the standardized innovation on future log-volatility.

The empirical implementation in this thesis is based on the Python `arch` package. In the notation of the `arch` package, the EGARCH model has p symmetric innovation terms, o asymmetric innovation terms and q lagged transformed conditional variance terms. Therefore, the EGARCH(1,1) model estimated in this thesis is implemented as an EGARCH specification with $p = 1$, $o = 1$ and $q = 1$. The corresponding variance recursion is

$$\log(\sigma_t^2) = \omega + \alpha_1 \left(|\varepsilon_{t-1}| - \sqrt{2/\pi} \right) + \gamma_1 \varepsilon_{t-1} + \beta_1 \log(\sigma_{t-1}^2). \quad (14)$$

Equation (13) presents the general EGARCH form, while equation (14) corresponds to the specification used in the empirical implementation. The difference concerns the centring term in the magnitude component. The general expression uses $\mathbb{E}|\varepsilon_{t-1}|$, whereas the `arch` implementation uses the fixed centring term $\sqrt{2/\pi}$. The EGARCH parameter estimates reported in this thesis should therefore be interpreted according to the implemented specification in equation (14).

The logarithmic specification implies that the conditional variance is positive without imposing the non-negativity restrictions required in ARCH and GARCH models. Since $\log(\sigma_t^2)$ can take any real value, exponentiating the recursion ensures that σ_t^2 itself remains strictly positive.

The parameters in equation (14) have distinct interpretations. The parameter β_1 measures the persistence of log-volatility. The parameter α_1 captures the effect of the magnitude of the standardized innovation on future log-volatility. A larger value of α_1 implies that large shocks, regardless of sign, have a stronger effect on the conditional variance.

The parameter γ_1 captures asymmetry in the volatility response. For two shocks of equal magnitude but opposite sign, $\varepsilon_{t-1} = c$ and $\varepsilon_{t-1} = -c$, with $c > 0$, the difference in

their contribution to log-volatility is determined by the term $\gamma_1 \varepsilon_{t-1}$. If $\gamma_1 < 0$, a negative shock increases future log-volatility more than a positive shock of the same magnitude. If $\gamma_1 > 0$, the opposite holds. In the context of Bitcoin, this parameter is interpreted as capturing asymmetric volatility responses or asymmetric news effects, rather than a leverage effect in the corporate finance sense.

For the EGARCH(1,1) model, a common condition for mean reversion in log-volatility is

$$|\beta_1| < 1.$$

Under this condition, shocks to log-volatility are not permanent. The EGARCH model is therefore useful for this thesis because it combines volatility persistence with the possibility that positive and negative Bitcoin return shocks affect future volatility differently. This allows the empirical analysis to examine whether **modelling** asymmetric volatility responses improves in-sample fit and out-of-sample forecasting performance relative to the symmetric ARCH and GARCH specifications.

4.3 Conditional innovation distributions

The ARCH, GARCH and EGARCH specifications describe the dynamics of the conditional variance σ_t^2 , but they do not by themselves fully determine the conditional distribution of returns. A complete model also requires an assumption on the distribution of the standardized innovation ε_t in the observation equation

$$y_t = \sigma_t \varepsilon_t.$$

Throughout the empirical analysis, the innovations are standardized. This standardization ensures that σ_t^2 can be interpreted as the conditional variance of y_t .

Let $f_\varepsilon(\cdot)$ denote the density of the standardized innovation. The conditional density of the mean-adjusted return is then given by

$$f(y_t | Y^{t-1}) = \frac{1}{\sigma_t} f_\varepsilon\left(\frac{y_t}{\sigma_t}\right). \quad (15)$$

The choice of f_ε is therefore an important part of the model specification, especially for Bitcoin returns, which display heavy tails and deviations from normality.

The normal distribution is included as the benchmark innovation distribution. Its density is

$$f_\varepsilon(e) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{e^2}{2}\right), \quad -\infty < e < \infty. \quad (16)$$

Under this assumption,

$$y_t | Y^{t-1} \sim N(0, \sigma_t^2).$$

The normal distribution is simple and provides a useful benchmark, but it is restrictive because the conditional innovation distribution does not allow for excess kurtosis or skewness.

To account for heavier tails, the Student- t distribution is considered as an alternative to the normal distribution. In volatility **modelling**, the Student- t distribution is commonly used because it assigns more probability mass to extreme observations than the normal distribution (Bollerslev, 1987). The standardized Student- t density is

$$f_\varepsilon(e; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) \sqrt{(\nu-2)\pi}} \left(1 + \frac{e^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \quad \nu > 2, \quad (17)$$

where the specification of the Gamma function is included in Appendix A.3 and ν denotes the degrees-of-freedom parameter. Smaller values of ν imply heavier tails, while the distribution approaches the standard normal distribution as $\nu \rightarrow \infty$. The restriction $\nu > 2$ ensures that the variance exists and is equal to one under this standardized parameterisation.

The skewed Student- t distribution extends the Student- t distribution by allowing the conditional innovation distribution to be asymmetric. This thesis uses the standardized skewed Student- t density of Hansen (1994). Let $\nu > 2$ denote the degrees-of-freedom parameter and let $-1 < \lambda < 1$ denote the skewness parameter. Define

$$a = 4\lambda c \frac{\nu - 2}{\nu - 1}, \quad b = \sqrt{1 + 3\lambda^2 - a^2}, \quad c = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi(\nu-2)}\Gamma(\frac{\nu}{2})}.$$

Furthermore, define

$$I_e = \begin{cases} 1, & \text{if } be + a \geq 0, \\ -1, & \text{if } be + a < 0. \end{cases}$$

The density of the standardized skewed Student- t innovation is then

$$f_\varepsilon(e; \nu, \lambda) = bc \left[1 + \frac{1}{\nu - 2} \left(\frac{be + a}{1 + \lambda I_e} \right)^2 \right]^{-\frac{\nu+1}{2}}. \quad (18)$$

When $\lambda = 0$, the skewed Student- t distribution reduces to the symmetric Student- t distribution. Positive and negative values of λ allow for different forms of asymmetry in the innovation distribution. This is relevant for Bitcoin returns because the descriptive statistics in Section 3.3 indicate that the empirical return distribution is not perfectly symmetric.

Finally, the generalized error distribution is included as an alternative flexible distribution for the standardized innovations. The GED is symmetric, but it allows the peak and tail thickness of the innovation distribution to differ from those of the normal benchmark (Nelson, 1991). Let $\kappa > 0$ denote the shape parameter and define

$$c_\kappa = \left(\frac{\Gamma(1/\kappa)}{2^{2/\kappa}\Gamma(3/\kappa)} \right)^{1/2}.$$

The standardized GED density is

$$f_\varepsilon(e; \kappa) = \frac{\kappa}{2^{1+1/\kappa}c_\kappa\Gamma(1/\kappa)} \exp \left[-\frac{1}{2} \left| \frac{e}{c_\kappa} \right|^\kappa \right], \quad \kappa > 0. \quad (19)$$

For $\kappa = 2$, the GED reduces to the standard normal distribution. Values below two imply heavier tails than the normal distribution, while values above two imply thinner tails. The GED therefore provides a flexible symmetric alternative for modelling leptokurtic return innovations.

In the empirical analysis, each volatility specification is estimated under the different innovation distributions. Hence, model comparison is not only based on the choice between ARCH, GARCH and EGARCH dynamics, but also on the assumed conditional distribution of ε_t . This is important because a model may capture volatility clustering while still providing an inadequate description of the conditional tail behaviour of Bitcoin returns. The preferred specification is therefore selected by jointly considering the volatility model and the innovation distribution using the in-sample and out-of-sample evaluation criteria described in Section 4.6 and Section 4.7

4.4 Maximum likelihood estimation

The parameters of the ARCH, GARCH and EGARCH models are estimated by maximum likelihood. Maximum likelihood estimation is appropriate in this setting because each model specification defines the one-step-ahead conditional distribution of the observed return. The conditional variance equation determines the time-varying scale σ_t^2 , while the assumed innovation distribution determines the shape of the conditional density.

Let θ denote the full vector of unknown parameters. This vector contains the constant mean parameter μ , the parameters of the conditional variance equation, and, when applicable, the additional parameters of the innovation distribution. For example, for an ARCH(q) model with Student- t innovations,

$$\theta = (\mu, \omega, \alpha_1, \dots, \alpha_q, \nu)',$$

where ν denotes the degrees-of-freedom parameter. For a GARCH(1,1) model with normally distributed innovations,

$$\theta = (\mu, \omega, \alpha_1, \beta_1)'.$$

For the skewed Student- t distribution, the parameter vector also contains the skewness parameter, while for the GED it contains the shape parameter.

For a given value of θ , the mean-adjusted return is

$$y_t(\theta) = r_t - \mu.$$

The standardized innovation is therefore given by

$$\varepsilon_t(\theta) = \frac{y_t(\theta)}{\sigma_t(\theta)} = \frac{r_t - \mu}{\sigma_t(\theta)}. \quad (20)$$

The conditional variance $\sigma_t^2(\theta)$ is obtained recursively from the relevant ARCH, GARCH or EGARCH specification. For example, under the GARCH(1,1) model,

$$\sigma_t^2(\theta) = \omega + \alpha_1 y_{t-1}^2(\theta) + \beta_1 \sigma_{t-1}^2(\theta),$$

where $y_{t-1}(\theta) = r_{t-1} - \mu$. Thus, for every candidate value of θ , the model implies a complete sequence of conditional variances and standardized innovations.

Let R^{t-1} denote the information set generated by past observed raw returns,

$$R^{t-1} = \{r_{t-1}, r_{t-2}, \dots\}.$$

The joint density of the observed return sample can be factorized into a product of conditional densities:

$$p(r_1, \dots, r_T; \theta) = p(r_1; \theta) \prod_{t=2}^T p(r_t \mid r_{t-1}, \dots, r_1; \theta).$$

In ARCH-family models, initial observations and initial conditional variances are required to start the variance recursion. These initial values are treated as given. Let $\mathcal{T}$ denote the set of observations included in the likelihood after accounting for the required lags and initialization. The conditional likelihood used in this thesis is then

$$L_T(\theta) = \prod_{t \in \mathcal{T}} f(r_t \mid R^{t-1}; \theta). \quad (21)$$

This conditional likelihood focuses on the one-step-ahead conditional densities implied by the volatility model, rather than on the marginal density of the initial observations.

Let $f_\varepsilon(\cdot; \eta)$ denote the density of the standardized innovation, where η collects the distribution-specific parameters. For the normal distribution, η is empty. For the Student- t distribution, $\eta = \nu$. For the skewed Student- t distribution, η contains both the tail and skewness parameters, while for the GED it contains the shape parameter. Since the model implies

$$r_t = \mu + \sigma_t \varepsilon_t,$$

the conditional density of the observed return is

$$f(r_t | R^{t-1}; \theta) = \frac{1}{\sigma_t(\theta)} f_\varepsilon\left(\frac{r_t - \mu}{\sigma_t(\theta)}; \eta\right). \quad (22)$$

The factor $1/\sigma_t(\theta)$ appears because the standardized innovation density is transformed to the scale of the observed return.

It is convenient to work with the logarithm of the likelihood. Taking logs of (21) gives the conditional log-likelihood

$$\ell_T(\theta) = \sum_{t \in \mathcal{T}} \log f(r_t | R^{t-1}; \theta). \quad (23)$$

Using (22), this can be written as

$$\ell_T(\theta) = \sum_{t \in \mathcal{T}} [-\log \sigma_t(\theta) + \log f_\varepsilon(\varepsilon_t(\theta); \eta)], \quad (24)$$

where $\varepsilon_t(\theta)$ is defined in (20). The maximum likelihood estimator is defined as

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \ell_T(\theta), \quad (25)$$

where Θ denotes the admissible parameter space.

The parameter space imposes the restrictions required by the model specification. For ARCH and GARCH models, the restrictions $\omega > 0$, $\alpha_i \geq 0$ and $\beta_j \geq 0$ ensure that the conditional variance remains non-negative. In addition, stationarity restrictions such as $\sum_{i=1}^q \alpha_i < 1$ for ARCH(q) and $\alpha_1 + \beta_1 < 1$ for GARCH(1,1) are imposed to ensure a finite unconditional variance. For the EGARCH(1,1) model, positivity of the conditional variance is guaranteed by specifying the model in terms of $\log(\sigma_t^2)$. A common stationarity condition for EGARCH(1,1) is $|\beta_1| < 1$, which ensures mean reversion in log-volatility. The distributional parameters are also restricted so that the standardized innovation distribution is well defined.

For the conditionally normal benchmark, the log-likelihood contribution of observation t is

$$\ell_t(\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma_t^2(\theta) - \frac{1}{2} \left(\frac{r_t - \mu}{\sigma_t(\theta)} \right)^2. \quad (26)$$

Equivalently, using $\varepsilon_t(\theta) = (r_t - \mu)/\sigma_t(\theta)$,

$$\ell_t(\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma_t^2(\theta) - \frac{1}{2} \varepsilon_t^2(\theta).$$

For the Student- t , skewed Student- t and GED specifications, the normal innovation density is replaced by the corresponding standardized innovation density introduced in [Section 4.3](#).

The volatility recursion remains unchanged for a given ARCH-family specification, but the likelihood contribution changes because the density of ε_t changes.

The maximization problem generally has no closed-form solution because the conditional variance depends recursively on past observations and unknown parameters. Therefore, the log-likelihood is maximized numerically. The same estimation procedure is applied to every combination of volatility specification and innovation distribution. The initial conditional variance is set equal to the in-sample variance of the raw return series. To ensure that likelihood-based comparisons are made on the same basis, all models are evaluated over the same likelihood sample.

After estimation, the fitted conditional variances are obtained by evaluating the corresponding variance recursion at the maximum likelihood estimates:

$$\hat{\sigma}_t^2 = \sigma_t^2(\hat{\theta}).$$

The fitted mean-adjusted returns are

$$\hat{y}_t = r_t - \hat{\mu},$$

and the standardized residuals are computed as

$$\hat{\varepsilon}_t = \frac{\hat{y}_t}{\hat{\sigma}_t} = \frac{r_t - \hat{\mu}}{\hat{\sigma}_t}. \quad (27)$$

These standardized residuals are used in the diagnostic analysis to assess whether the fitted model has adequately captured the remaining serial dependence in volatility.

Under standard regularity conditions and correct specification of the conditional density, the maximum likelihood estimator is consistent and asymptotically normally distributed. Statistical inference is based on the observed information matrix. Let $H_T(\hat{\theta})$ denote the Hessian matrix of the total log-likelihood evaluated at the maximum:

$$H_T(\hat{\theta}) = \left. \frac{\partial^2 \ell_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\hat{\theta}}.$$

An estimate of the covariance matrix of the maximum likelihood estimator is then given by

$$\widehat{\text{Var}}(\hat{\theta}) = \left[-H_T(\hat{\theta}) \right]^{-1}. \quad (28)$$

The standard errors are obtained by taking the square roots of the diagonal elements of this covariance matrix.

4.5 Model diagnostics

After estimating the ARCH, GARCH and EGARCH models, diagnostic checks are used to assess whether the fitted models adequately capture the remaining dependence structure in the return series. The diagnostics are based on the standardized residuals obtained after maximum likelihood estimation,

$$\hat{\varepsilon}_t = \frac{\hat{y}_t}{\hat{\sigma}_t} = \frac{r_t - \hat{\mu}}{\hat{\sigma}_t}.$$

If the model is correctly specified, the standardized residuals should behave approximately like independent standardized innovations. In particular, they should display no remaining serial correlation, and their squared values should not exhibit systematic dependence.

The first diagnostic check concerns serial correlation in the standardized residuals. The autocorrelation function of $\hat{\varepsilon}_t$ is inspected, and the Ljung–Box test is applied to test whether the standardized residuals are jointly uncorrelated up to selected lag lengths. Remaining autocorrelation would indicate that the model has not fully captured the dynamic structure in the conditional mean.

The second diagnostic check concerns remaining dependence in volatility. For this purpose, the autocorrelation function of the squared standardized residuals, $\hat{\varepsilon}_t^2$, is inspected and the Ljung–Box test is applied to $\hat{\varepsilon}_t^2$. If the volatility model is adequate, the squared standardized residuals should no longer display the serial dependence that was present in the squared raw returns.

Finally, the ARCH–LM test is applied to the standardized residuals to test directly for remaining ARCH effects. The null hypothesis is that there are no remaining ARCH effects. Rejection of this null hypothesis would suggest that the fitted volatility model has not fully captured the conditional heteroskedasticity in the return series.

For comparability across models, the same diagnostic tests and lag lengths are used for each estimated specification. In line with the preliminary analysis in [Section 3.5](#), the Ljung–Box tests are reported at lags 10, 20 and 30, while the ARCH–LM test is applied using 10 lags. The diagnostic results are interpreted as model adequacy checks and are considered together with the in-sample comparison criteria and the out-of-sample forecast evaluation discussed in the following subsections.

4.6 In-sample model comparison

After estimating the candidate volatility models, their in-sample performance is compared using likelihood-based information criteria. This comparison is carried out at the level of the complete model specification. Hence, a model is defined not only by its volatility dynamics, such as ARCH, GARCH or EGARCH, but also by its assumed innovation distribution. For example, a GARCH(1,1) model with Student- t innovations and an EGARCH(1,1) model with skewed Student- t innovations are treated as distinct specifications.

A natural measure of in-sample fit is the maximized log-likelihood. For model m , let $\hat{\theta}_m$ denote the maximum likelihood estimate and let

$$\ell_m(\hat{\theta}_m)$$

denote the total maximized conditional log-likelihood, as defined in [Section 4.4](#). A higher value of the maximized log-likelihood indicates that the model assigns a higher probability to the observed return sequence. However, the maximized log-likelihood alone is not sufficient for model comparison, because models with more parameters can often achieve a better fit simply by being more flexible. Therefore, information criteria are used to balance goodness of fit against model complexity.

The first criterion is the Akaike Information Criterion (AIC), introduced by Akaike (1974). For model m , it is defined as

$$\text{AIC}_m = -2\ell_m(\hat{\theta}_m) + 2k_m, \quad (29)$$

where k_m denotes the number of estimated parameters in model m . The first term rewards in-sample fit through the maximized log-likelihood, while the second term penalizes the number of parameters. A lower AIC value indicates a preferred model according to this criterion.

The second criterion is the Bayesian Information Criterion (BIC), introduced by Schwarz (1978). It is defined as

$$\text{BIC}_m = -2\ell_m(\hat{\theta}_m) + k_m \log(n_\ell), \quad (30)$$

where n_ℓ denotes the number of observations included in the likelihood evaluation. As with the AIC, lower BIC values indicate better model performance. The BIC imposes a stronger penalty for additional parameters than the AIC when the sample size is sufficiently large. Consequently, the BIC tends to **favour** more parsimonious specifications.

In this thesis, the parameter count k_m includes all parameters estimated by maximum likelihood. This includes the constant mean parameter μ , the parameters of the conditional variance equation, and any additional parameters of the innovation distribution. For example, a GARCH(1,1) model with normally distributed innovations contains the parameters

$$(\mu, \omega, \alpha_1, \beta_1),$$

so that $k_m = 4$. A GARCH(1,1) model with Student- t innovations additionally includes the degrees-of-freedom parameter ν , giving $k_m = 5$.

For comparability, all information criteria are computed using the same likelihood sample. This is important because ARCH, GARCH and EGARCH specifications require initial observations to start their recursions, and likelihood values should not be compared across models evaluated on different numbers of observations. The total maximized log-likelihood is used when computing AIC and BIC.

The AIC and BIC provide in-sample evidence on how well each specification balances fit and complexity. They are therefore useful for identifying which volatility model and innovation distribution provide the best statistical description of the observed Bitcoin return series. However, a model with superior in-sample performance does not necessarily produce the most accurate volatility forecasts. For this reason, the in-sample comparison is complemented by the out-of-sample forecast evaluation described in **Section 4.7**.

4.7 Forecasting procedure and out-of-sample evaluation

The final step of the empirical analysis evaluates the out-of-sample forecasting performance of the estimated volatility models. While the in-sample criteria discussed in **Section 4.6** assess how well a model describes the observed return series after penalizing for model complexity, out-of-sample evaluation examines whether the model produces accurate volatility forecasts for observations not used in estimation.

The return sample is divided into an in-sample period and an out-of-sample period. The first 80% of the return observations are used for estimation and the remaining 20% are reserved for forecast evaluation. Let T_0 denote the final observation in the in-sample period. The models are estimated using observations up to time T_0 , while the remaining observations are reserved for forecast evaluation. The out-of-sample evaluation period is therefore

$$t = T_0 + 1, \dots, T.$$

Let

$$H = T - T_0$$

denote the number of out-of-sample observations.

For each model m , the parameter vector θ_m is estimated by maximum likelihood using the in-sample observations. The resulting estimate is denoted by $\hat{\theta}_m$. The estimated parameters are then kept fixed during the out-of-sample evaluation. For each day in the

out-of-sample period, a one-step-ahead forecast of the conditional variance is constructed using information available up to the previous day. This forecast is denoted by

$$\hat{\sigma}_{t|t-1,m}^2,$$

where the subscript $t|t-1$ indicates that the variance for day t is forecast using information available at time $t-1$.

The one-step-ahead forecast is obtained from the fitted volatility recursion. For example, for a GARCH(1,1) model the forecast is given by

$$\hat{\sigma}_{t|t-1,m}^2 = \hat{\omega}_m + \hat{\alpha}_{1,m}\hat{y}_{t-1,m}^2 + \hat{\beta}_{1,m}\hat{\sigma}_{t-1|t-2,m}^2,$$

where

$$\hat{y}_{t-1,m} = r_{t-1} - \hat{\mu}_m.$$

Analogous recursions are used for the ARCH and EGARCH specifications. In each case, the forecast is based only on information that would have been available before observing r_t .

A complication in volatility forecast evaluation is that the true conditional variance is not directly observed. Therefore, an observable proxy for volatility is required. Since this thesis uses daily return data, the squared mean-adjusted return is used as the volatility proxy. To keep the **realised** proxy identical across models, the out-of-sample proxy is constructed using the in-sample mean of the raw return series. Let $\bar{r}_{\text{in}}$ denote the in-sample average of r_t . The **realised** volatility proxy is then defined as

$$v_t = (r_t - \bar{r}_{\text{in}})^2.$$

This proxy is noisy because a single squared daily return also reflects the **realised** innovation in that period. Nevertheless, squared returns are commonly used as volatility proxies when only daily data are available, although the literature emphasizes that they are imperfect measures of the latent conditional variance (Patton, 2011).

The forecast accuracy of model m is evaluated by comparing $\hat{\sigma}_{t|t-1,m}^2$ with v_t over the out-of-sample period. The first loss measure is the root mean squared error,

$$\text{RMSE}_m = \sqrt{\frac{1}{H} \sum_{t=T_0+1}^T \left(v_t - \hat{\sigma}_{t|t-1,m}^2 \right)^2}. \quad (31)$$

The RMSE penalizes large forecast errors relatively strongly because the errors are squared before averaging.

The second loss measure is the mean absolute error,

$$\text{MAE}_m = \frac{1}{H} \sum_{t=T_0+1}^T \left| v_t - \hat{\sigma}_{t|t-1,m}^2 \right|. \quad (32)$$

The MAE is less sensitive to extreme forecast errors than the RMSE and therefore provides a complementary measure of forecast accuracy.

In addition, this thesis reports the quasi-likelihood loss function, which is frequently used in volatility forecast evaluation when the conditional variance is the object of interest (Patton, 2011). For model m , it is defined as

$$\text{QLIKE}_m = \frac{1}{H} \sum_{t=T_0+1}^T \left[\log \left(\hat{\sigma}_{t|t-1,m}^2 \right) + \frac{v_t}{\hat{\sigma}_{t|t-1,m}^2} \right]. \quad (33)$$

The QLIKE criterion is useful because it evaluates the forecast as a variance forecast and penalizes cases where the predicted variance is too low relative to the realised squared return.

For all three loss functions, lower values indicate better out-of-sample volatility forecast performance. The forecast evaluation is conducted over the same out-of-sample period for all specifications. This ensures that differences in forecast performance can be attributed to the model specification rather than to differences in the evaluation sample.

The out-of-sample results are interpreted together with the diagnostic tests and the in-sample information criteria. A model that performs well in-sample does not necessarily provide the most accurate volatility forecasts. Conversely, a model with slightly weaker in-sample fit may still be preferable if it produces more accurate out-of-sample volatility forecasts. The final model assessment therefore considers both the statistical fit of the model and its forecasting performance.

5 Empirical Results

5.1 In-sample model comparison

This subsection reports the in-sample comparison of the estimated volatility models. As discussed in Section 4, each candidate specification is defined by both a conditional variance model and an innovation distribution. The empirical analysis therefore compares ARCH, GARCH and EGARCH specifications jointly with the normal, Student-t, GED and skewed Student-t innovation distributions. All models are estimated over the same in-sample period, consisting of 3,042 daily Bitcoin log return observations, so that the likelihood-based criteria are directly comparable across specifications.

Table 7 reports the top 5 estimated models according to the Akaike Information Criterion. The results show that the EGARCH(1,1) model with Student-t innovations provides the lowest AIC value. This model has a maximized log-likelihood of -7609.1981 , an AIC value of 15230.3963 and a BIC value of 15266.5179 . The same model is also selected by the BIC, as shown in Appendix A.4.2. Hence, both likelihood-based information criteria identify the EGARCH(1,1) model with Student-t innovations as the preferred in-sample specification.

Table 7: AIC ranking of top 5 estimated volatility models.

Model	Distribution	Log-likelihood	AIC	BIC	# Parameters
EGARCH(1,1)	Student-t	-7609.20	15230.40	15266.52	6
EGARCH(1,1)	Skewed Student-t	-7609.13	15232.26	15274.40	7
EGARCH(1,1)	GED	-7627.92	15267.84	15303.96	6
GARCH(1,1)	Student-t	-7638.87	15287.74	15317.84	5
GARCH(1,1)	GED	-7639.46	15288.91	15319.01	5

The ranking provides clear evidence that both the volatility specification and the innovation distribution matter for describing Bitcoin return volatility. The three best models according to AIC are all EGARCH specifications with non-normal innovations. Specifically, the EGARCH(1,1) model with Student-t innovations ranks first, followed by the EGARCH(1,1) model with skewed Student-t innovations and the EGARCH(1,1) model with GED innovations. This suggests that allowing for asymmetric volatility responses

improves the in-sample description of the return series relative to the symmetric ARCH and GARCH alternatives.

The results also show that heavy-tailed innovation distributions are important. The highest-ranked specifications almost all use Student-t, GED or skewed Student-t innovations, whereas the models with normally distributed innovations perform substantially worse. For example, the EGARCH(1,1) model with normal innovations ranks only sixteenth according to both AIC and BIC, despite having the same volatility structure as the best-ranked EGARCH specification. This pattern is consistent with the descriptive statistics in [Section 3.3](#), which showed strong excess kurtosis and a clear rejection of normality for the raw Bitcoin return series.

An additional observation is that the EGARCH(1,1) model with skewed Student-t innovations attains a slightly higher log-likelihood than the EGARCH(1,1) model with Student-t innovations. Its maximized log-likelihood equals -7609.1282 , compared with -7609.1981 for the Student-t specification. However, the skewed Student-t model includes one additional distributional parameter. After penalizing for this additional parameter, both AIC and BIC prefer the simpler Student-t specification. This illustrates that the improvement in likelihood from allowing skewness in the innovation distribution is not large enough to compensate for the increase in model complexity.

The ARCH specifications perform less well than the GARCH and EGARCH specifications in terms of likelihood-based fit. Among the ARCH models, the ARCH(3) model with GED innovations performs best, ranking seventh according to both AIC and BIC. Lower-order ARCH models perform worse, which suggests that several lagged squared shocks are needed to capture part of the volatility dependence in Bitcoin returns. Nevertheless, the GARCH(1,1) and EGARCH(1,1) specifications achieve better in-sample fit with a more parsimonious dynamic structure, supporting the relevance of [modelling](#) volatility persistence through lagged conditional variances.

Overall, the in-sample evidence indicates that the best statistical description of daily Bitcoin return volatility is obtained by combining an asymmetric volatility model with a heavy-tailed innovation distribution. The EGARCH(1,1) model with Student-t innovations is therefore selected as the preferred in-sample model according to both AIC and BIC. This result supports the empirical relevance of volatility persistence, asymmetric volatility responses and heavy-tailed conditional innovations. However, likelihood-based in-sample fit does not necessarily imply superior forecasting performance. The out-of-sample forecast evaluation in [Section 6](#) therefore examines whether the same specification also provides the most accurate volatility forecasts.

5.2 Parameter estimates of the selected in-sample model

Since both AIC and BIC select the EGARCH(1,1) model with Student-t innovations, this subsection discusses the parameter estimates of this specification in more detail. The estimates are reported in [Table 8](#).

Table 8: Parameter estimates for the EGARCH(1,1) model with Student-t innovations.

Parameter	Estimate	std_error	t_value	p_value
mu	0.1346	0.0343	3.9215	0.0001
omega	0.0842	0.0246	3.4170	0.0006
alpha[1]	0.2580	0.0352	7.3286	< 0.0000
gamma[1]	0.0394	0.0155	2.5355	0.0112
beta[1]	0.9944	0.0039	257.7361	< 0.0000
nu	2.5229	0.1495	16.8732	< 0.0000

The EGARCH intercept is estimated as $\hat{\omega} = 0.0842$ and is statistically significant. In the EGARCH specification, this parameter enters the recursion for $\log(\sigma_t^2)$ rather than the conditional variance directly. It should therefore be interpreted as an intercept in the log-variance equation, not as a long-run variance level in the same way as the intercept in an ARCH or GARCH model.

The estimate of the magnitude parameter is $\hat{\alpha}_1 = 0.2580$. This parameter is positive and statistically significant. It measures the effect of the absolute standardized innovation on future log-volatility. Hence, large standardized shocks increase subsequent volatility, independently of whether the shock is positive or negative. This result is consistent with the volatility clustering documented in [Section 3.5](#).

The persistence parameter is estimated as $\hat{\beta}_1 = 0.9944$. This value is very close to one, indicating highly persistent log-volatility. At the same time, the estimate satisfies the common EGARCH mean-reversion condition $|\beta_1| < 1$. This suggests that volatility shocks are not permanent, but that they decay only slowly over time. The high degree of persistence is consistent with the empirical [behaviour](#) of Bitcoin returns, where periods of high volatility tend to persist for several days or longer.

The asymmetry parameter is estimated as $\hat{\gamma}_1 = 0.0394$ and is statistically significant. Under the EGARCH specification used in this thesis, a positive value of γ_1 implies that positive standardized innovations increase future log-volatility more than negative standardized innovations of the same magnitude. This finding indicates the presence of asymmetric volatility responses in Bitcoin returns.

Finally, the estimated Student-t degrees-of-freedom parameter is $\hat{\nu} = 2.5229$. This value is relatively low and indicates very heavy-tailed standardized innovations. The result is consistent with the descriptive statistics in [Section 3.3](#), where the raw Bitcoin return distribution was found to be strongly leptokurtic and the Jarque–Bera test rejected normality. The parameter estimate therefore confirms that heavy-tailed innovations are important even after accounting for time-varying volatility.

5.3 Residual diagnostics

After comparing the models using likelihood-based information criteria, the next step is to examine whether the fitted specifications adequately capture the remaining dependence structure in the return series. As discussed in [Section 4.5](#), the diagnostics are based on the standardized residuals $\hat{\varepsilon}_t$. If a volatility model is correctly specified, the standardized residuals should display no remaining serial correlation, and the squared standardized residuals should not exhibit systematic dependence. The diagnostic analysis therefore considers the Ljung–Box test applied to $\hat{\varepsilon}_t$, the Ljung–Box test applied to $\hat{\varepsilon}_t^2$, and the ARCH–LM test applied to the standardized residuals.

The diagnostic results show that the best in-sample model, the EGARCH(1,1) specification with Student-t innovations, does not pass all diagnostic checks at the 5% significance level. The Ljung–Box p-value for the standardized residuals equals 0.0056, indicating rejection of the null hypothesis of no residual autocorrelation. However, the same model performs well according to the volatility-focused diagnostics. The Ljung–Box p-value for the squared standardized residuals equals 0.6791, and the ARCH–LM p-value equals 0.6882. Hence, there is little evidence of remaining serial dependence in the squared standardized residuals or of remaining ARCH effects. This suggests that the selected EGARCH model captures the main conditional variance dynamics, although some linear dependence remains in the standardized residuals.

Table 9: Model diagnostics of EGARCH(1,1).

Model	Distribution	LB P-Value (standardized residuals)	LB P-Value (Squared Standardized Residuals)	LM P-Value
EGARCH(1,1)	Student-t	0.01	0.68	0.69

A similar pattern is observed for the highest-ranked GARCH specifications. For example, the GARCH(1,1) model with GED innovations has a Ljung–Box p-value of 0.0141 for the standardized residuals, but much larger p-values for the squared standardized residuals and ARCH–LM test, equal to 0.8980 and 0.8978, respectively. This indicates that, while some residual autocorrelation remains in the standardized residuals, the model removes most of the volatility dependence that was present in the raw return series. This distinction is important because the objective of this thesis is to model and forecast conditional volatility rather than to construct a detailed dynamic model for the conditional mean.

Table 10 reports the specifications that pass all three diagnostic checks at the 5% level. Only three models satisfy this criterion: ARCH(3) with GED innovations, ARCH(3) with Student-t innovations, and ARCH(3) with skewed Student-t innovations. Among these, ARCH(3) with GED innovations has the lowest AIC and BIC values and is therefore the preferred model within the subset of diagnostically adequate specifications.

Table 10: Diagnostically adequate models for estimated volatility models.

Model	Distribution	LB P-Value (standardized residuals)	LB P-Value (Squared Standardized Residuals)	LM P-Value
ARCH(3)	GED	0.22	0.23	0.22
ARCH(3)	Student-t	0.22	0.37	0.35
ARCH(3)	Skewed Student-t	0.22	0.37	0.35

The fact that only ARCH(3) specifications pass all diagnostic checks is noteworthy. These models are not the best specifications according to AIC and BIC, but they produce standardized residuals with less evidence of remaining serial dependence according to the diagnostic summary. This shows that model selection based on likelihood-based fit and model adequacy based on residual diagnostics do not necessarily lead to the same ranking. In particular, the EGARCH(1,1) model with Student-t innovations provides the best in-sample likelihood-based fit, whereas the ARCH(3) model with GED innovations provides the strongest residual diagnostic performance among the models considered.

The diagnostic results should therefore be interpreted with some nuance. The best in-sample EGARCH and GARCH specifications do not fully remove autocorrelation in the standardized residuals. However, they perform well in the diagnostics that are most directly related to volatility dependence, namely the tests on squared standardized residuals and remaining ARCH effects. This suggests that these models capture the main time-varying variance dynamics of Bitcoin returns, even though the conditional mean specification may remain somewhat simplified. Since the empirical focus of this thesis is volatility **modelling**, the remaining autocorrelation in standardized residuals is treated as a limitation rather than as evidence that the volatility models are unsuitable.

6 Forecast Evaluation and Model Comparison

The previous section evaluated the estimated volatility models using in-sample likelihood criteria and residual diagnostics. This section assesses whether the same models also produce accurate out-of-sample volatility forecasts. As discussed in **Section** 4.7, the models are estimated using the in-sample period, while the remaining observations are reserved for forecast evaluation. For each model, one-step-ahead forecasts of the conditional variance are compared with the **realised** volatility proxy constructed from squared mean-adjusted returns.

6.1 Out-of-sample forecast accuracy

Table 11 reports the out-of-sample forecast accuracy of the top 5 estimated models based on RMSE. Forecast performance is evaluated using RMSE, MAE and QLIKE.

Table 11: Top 5 out-of-sample volatility forecast ranking by RMSE.

Model	Distribution	RMSE	MAE	QLIKE
GARCH(1,1)	GED	14.63	7.39	2.70
EGARCH(1,1)	GED	14.64	7.66	2.72
GARCH(1,1)	Student-t	14.69	7.50	2.71
GARCH(1,1)	Skewed Student-t	14.69	7.50	2.71
EGARCH(1,1)	Normal	14.71	8.73	2.77

The results show that the GARCH(1,1) model with GED innovations provides the best out-of-sample performance according to all three loss functions. The fact that the same specification is selected by all three forecast evaluation criteria strengthens the conclusion that this model provides the most accurate variance forecasts over the out-of-sample period.

This result is notable because the GARCH(1,1) model with GED innovations was not the preferred model according to the in-sample likelihood criteria. In **Section** 5.1, the EGARCH(1,1) model with Student-t innovations was selected by both AIC and BIC. However, in the out-of-sample forecast evaluation, this model performs substantially less well, ranking thirteenth according to RMSE, twelfth according to MAE and tenth according to QLIKE. Hence, the model with the strongest in-sample fit does not provide the most accurate out-of-sample volatility forecasts.

The forecast rankings also confirm the importance of non-normal innovation distributions. The highest-ranked specifications mostly use GED, Student-t or skewed Student-t

innovations rather than normal innovations. This is consistent with the descriptive evidence in Section 3.3, where Bitcoin returns were found to be strongly heavy-tailed. Allowing for heavier-tailed conditional innovations therefore improves not only the in-sample likelihood fit, but also the out-of-sample forecast performance.

At the same time, the results suggest that adding asymmetry through the EGARCH specification does not necessarily improve volatility forecasting in this sample. Although EGARCH specifications perform well according to AIC and BIC, the symmetric GARCH(1,1) model with GED innovations gives the best out-of-sample forecasts. This indicates that, for the forecast period considered here, a parsimonious persistent volatility model with a flexible heavy-tailed innovation distribution performs better than the more flexible asymmetric EGARCH specification.

6.2 Comparison of selected volatility forecasts

The forecast accuracy rankings are further illustrated by the out-of-sample forecast plots. Figure 5 compares the realised volatility proxy with the conditional variance forecasts of selected models. The realised volatility proxy is highly variable and contains several large spikes. This behaviour is expected because the proxy is based on a single squared daily return and is therefore a noisy measure of the latent conditional variance. The model-based forecasts are much smoother, reflecting the fact that ARCH-family models forecast the conditional variance rather than the realised squared return itself.

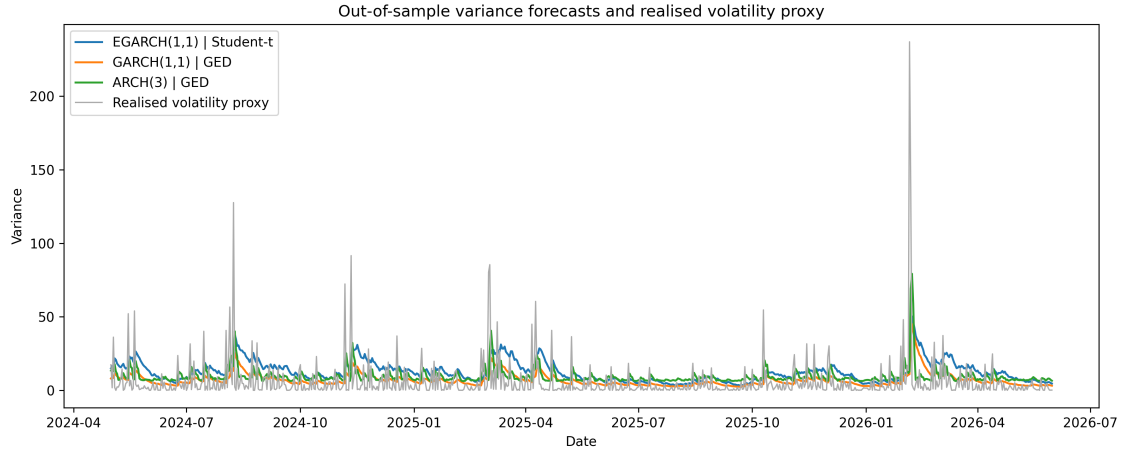


Figure 5: Out-of-sample realised volatility proxy and one-step-ahead conditional variance forecasts for selected models.

The largest realised volatility spikes are not fully anticipated by any of the models. This is not surprising. A one-step-ahead volatility forecast can only use information available up to the previous day. Therefore, extreme return realisations are difficult to predict before they occur. The models can respond to such shocks by increasing subsequent volatility forecasts, but they cannot generally forecast the exact timing and magnitude of isolated extreme events.

Figure 6 removes the realised volatility proxy and compares only the selected model forecasts. This makes the differences between the volatility forecasts clearer. The GARCH(1,1) model with GED innovations generally produces smoother and more moderate forecasts than the EGARCH(1,1) model with Student-t innovations. The EGARCH specification often produces higher variance forecasts, especially during volatile periods. This helps ex-

plain why the GARCH(1,1)-GED model performs better according to MAE and QLIKE: it avoids overpredicting volatility during calmer parts of the evaluation period while still responding to changes in market conditions.

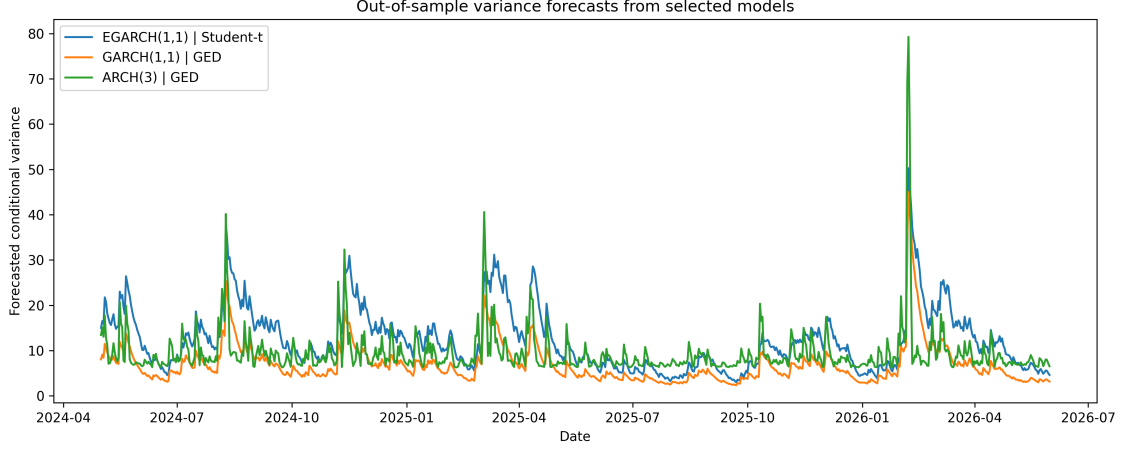


Figure 6: Out-of-sample one-step-ahead conditional variance forecasts for selected models.

The ARCH(3) model with GED innovations behaves differently from the GARCH and EGARCH specifications. Its forecasts react more directly to recent squared shocks and are therefore more irregular. This is consistent with the structure of ARCH models, where the conditional variance depends only on lagged squared returns and not on lagged conditional variances. Although the ARCH(3)-GED model performed well in the residual diagnostics of [Section 5.3](#), its out-of-sample forecasts are less accurate than those of the GARCH(1,1)-GED model.

The cumulative forecast error plots provide additional insight into the ranking of the selected models. Figure 7 shows the cumulative squared forecast errors. Since the RMSE is based on squared errors, this figure indicates when the RMSE differences between models arise. The cumulative squared error increases sharply around periods with large [realised](#) volatility shocks, especially when the [realised](#) volatility proxy displays extreme spikes. This confirms that RMSE is strongly influenced by large forecast errors.

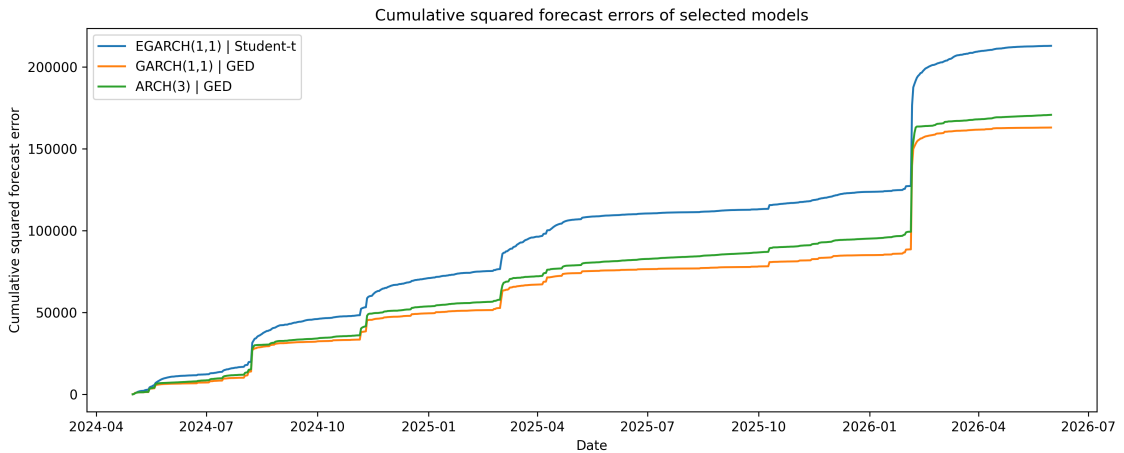


Figure 7: Cumulative squared forecast errors for selected volatility models over the out-of-sample period.

Figure 8 reports the cumulative absolute forecast errors. The GARCH(1,1) model with GED innovations remains below the other selected models over most of the evaluation period. This supports the MAE ranking and indicates that the superior forecast performance of the GARCH(1,1)-GED model is not driven solely by a few extreme squared errors. Instead, the model provides more accurate forecasts on average throughout the out-of-sample period.

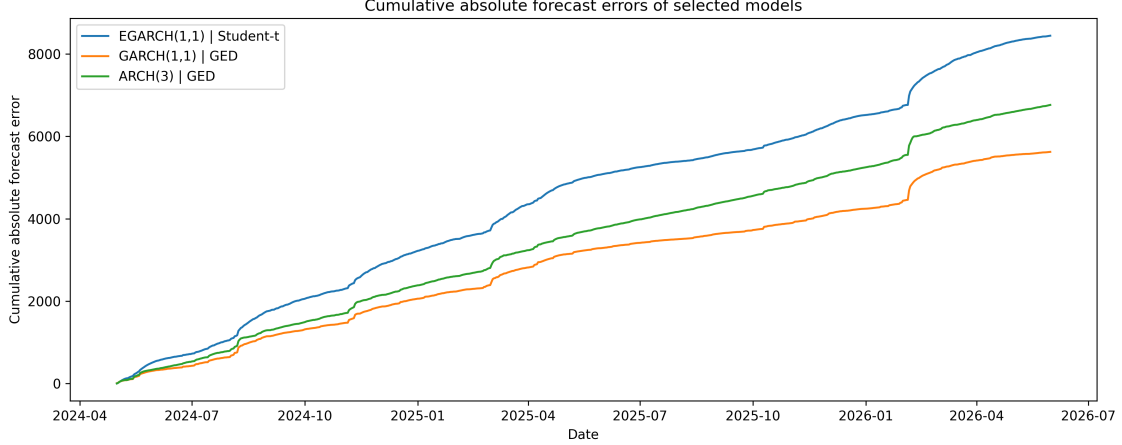


Figure 8: Cumulative absolute forecast errors for selected volatility models over the out-of-sample period.

6.3 Overall model comparison

Table 12 combines the in-sample rankings, diagnostic results and out-of-sample forecast rankings. The table highlights that the different evaluation criteria do not select the same model. The EGARCH(1,1) model with Student-t innovations ranks first according to both AIC and BIC, while the GARCH(1,1) model with GED innovations ranks first according to RMSE, MAE and QLIKE. The ARCH(3) model with GED innovations is the preferred specification among the models that pass all residual diagnostic checks.

Table 12: Top 5 models evaluation summary ranked by RMSE.

Model	Distribution	AIC	BIC	RMSE	MAE	QLIKE	# Parameters
GARCH(1,1)	GED	15288.91	15319.01	14.63	7.39	2.70	5
EGARCH(1,1)	GED	15267.84	15303.96	14.64	7.66	2.72	6
GARCH(1,1)	Student-t	15287.74	15317.84	14.69	7.50	2.71	5
GARCH(1,1)	Skewed Student-t	15289.72	15325.84	14.69	7.50	2.71	6
EGARCH(1,1)	Normal	16089.43	16119.53	14.71	8.73	2.77	5

This distinction is important for answering the research question of this thesis. If the objective is to obtain the best in-sample statistical description of daily Bitcoin returns, the results favour the EGARCH(1,1) model with Student-t innovations. This model captures volatility persistence, asymmetric volatility responses and heavy-tailed conditional innovations. However, if the objective is to forecast conditional variance out of sample, the GARCH(1,1) model with GED innovations performs best. This model combines volatility persistence with a flexible heavy-tailed innovation distribution, while remaining more parsimonious than the EGARCH alternatives.

The results therefore show that model fit, diagnostic adequacy and forecast accuracy are related but distinct dimensions of model performance. A model that fits the in-sample conditional distribution well does not necessarily produce the most accurate volatility forecasts. Similarly, a model that passes all residual diagnostics does not necessarily minimize forecast loss. For this reason, the final model assessment should not be based on a single criterion.

Overall, the out-of-sample evidence supports the GARCH(1,1) model with GED innovations as the preferred forecasting model for daily Bitcoin return volatility. This conclusion is robust across RMSE, MAE and QLIKE. At the same time, the in-sample results from [Section 5.1](#) indicate that EGARCH(1,1) with Student-t innovations provides the strongest likelihood-based fit. The empirical findings therefore suggest a nuanced answer to the research question: asymmetric EGARCH dynamics are useful for describing the in-sample distribution of Bitcoin returns, but the most accurate out-of-sample volatility forecasts are obtained from a symmetric GARCH(1,1) model with GED innovations.

7 Discussion

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8 Conclusion

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9 Future Research

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A Formulas and tables

A.1 GARCH(1,1) unfolding

The ARCH(∞) representation of a GARCH(1,1) model is obtained unfolding the GARCH updating equation as follows

Proof.

$$\begin{aligned}
\sigma_t^2 &= \omega + \beta_1 \sigma_{t-1}^2 + \alpha_1 y_{t-1}^2 \\
&= \omega + \beta_1 (\omega + \beta_1 \sigma_{t-2}^2 + \alpha_1 y_{t-2}^2) + \alpha_1 y_{t-1}^2 \\
&= \omega + \beta_1 \omega + \beta_1^2 \sigma_{t-2}^2 + \alpha_1 \beta_1 y_{t-2}^2 + \alpha_1 y_{t-1}^2 \\
&= \omega + \beta_1 \omega + \beta_1^2 (\omega + \beta_1 \sigma_{t-3}^2 + \alpha_1 y_{t-3}^2) + \alpha_1 \beta_1 y_{t-2}^2 + \alpha_1 y_{t-1}^2 \\
&= \omega + \beta_1 \omega + \beta_1^2 \omega + \beta_1^3 \sigma_{t-3}^2 + \alpha_1 \beta_1^2 y_{t-3}^2 + \alpha_1 \beta_1 y_{t-2}^2 + \alpha_1 y_{t-1}^2.
\end{aligned}$$

Therefore, after k iterations we get

$$\begin{aligned}
\sigma_t^2 &= \omega + \dots + \beta_1^{k-1} \omega + \beta_1^k \sigma_{t-k}^2 + \alpha_1 \beta_1^{k-1} y_{t-k}^2 + \dots + \alpha_1 \beta_1 y_{t-2}^2 + \alpha_1 y_{t-1}^2 \\
&= \omega \sum_{i=0}^{k-1} \beta_1^i + \beta_1^k \sigma_{t-k}^2 + \alpha_1 \sum_{i=0}^{k-1} \beta_1^i y_{t-i-1}^2.
\end{aligned}$$

Now we can take the limit $k \rightarrow \infty$ and obtain

$$\begin{aligned}
\sigma_t^2 &= \omega \sum_{i=0}^{\infty} \beta_1^i + \alpha_1 \sum_{i=0}^{\infty} \beta_1^i y_{t-i-1}^2 \\
&= \frac{\omega}{1 - \beta_1} + \alpha_1 \sum_{i=0}^{\infty} \beta_1^i y_{t-i-1}^2.
\end{aligned}$$

□

This representation shows that the GARCH(1,1) model incorporates information from many past squared innovations, but with geometrically declining weights. Compared with an ARCH model with many lags, the GARCH model can therefore capture long-lasting volatility dependence with only a small number of parameters.

A.2 Robustness check ADF

Table 13: ADF test results for alternative fixed lag lengths

Fixed lag length	ADF statistic	p-value	Conclusion
0	-63.5751	$< 10^{-16}$	Reject unit root
1	-42.9431	$< 10^{-16}$	Reject unit root
2	-35.1561	$< 10^{-16}$	Reject unit root
5	-23.8215	$< 10^{-16}$	Reject unit root
10	-17.6401	3.76×10^{-30}	Reject unit root
20	-12.3974	4.64×10^{-23}	Reject unit root
30	-10.2648	4.15×10^{-18}	Reject unit root

A.3 Gamma function

The Student- t and skewed Student- t distribution make use of the Gamma function. For $x > 0$, the Gamma function is defined as

$$\Gamma(x) = \int_0^{\infty} u^{x-1} e^{-u} du.$$

It satisfies the recursion

$$\Gamma(x+1) = x\Gamma(x),$$

and for positive integers n this implies $\Gamma(n) = (n-1)!$. The Gamma function can therefore be viewed as a continuous extension of the factorial function.

A.4 Empirical result tables

A.4.1 AIC Ranking table

Table 14: Full AIC ranking of estimated volatility models.

Model	Distribution	Log-likelihood	AIC	BIC	# Parameters
EGARCH(1,1)	Student-t	-7609.20	15230.40	15266.52	6
EGARCH(1,1)	Skewed Student-t	-7609.13	15232.26	15274.40	7
EGARCH(1,1)	GED	-7627.92	15267.84	15303.96	6
GARCH(1,1)	Student-t	-7638.87	15287.74	15317.84	5
GARCH(1,1)	GED	-7639.46	15288.91	15319.01	5
GARCH(1,1)	Skewed Student-t	-7638.86	15289.72	15325.84	6
ARCH(3)	GED	-7747.68	15507.36	15543.48	6
ARCH(3)	Student-t	-7756.58	15525.16	15561.28	6
ARCH(3)	Skewed Student-t	-7756.58	15527.15	15569.29	7
ARCH(2)	GED	-7767.12	15544.24	15574.34	5
ARCH(2)	Student-t	-7778.16	15566.31	15596.42	5
ARCH(2)	Skewed Student-t	-7778.14	15568.29	15604.41	6
ARCH(1)	GED	-7795.42	15598.85	15622.93	4
ARCH(1)	Student-t	-7807.70	15623.40	15647.48	4
ARCH(1)	Skewed Student-t	-7807.69	15625.37	15655.48	5
EGARCH(1,1)	Normal	-8039.71	16089.43	16119.53	5
GARCH(1,1)	Normal	-8051.16	16110.32	16134.40	4
ARCH(3)	Normal	-8208.08	16426.16	16456.26	5
ARCH(2)	Normal	-8231.13	16470.26	16494.34	4
ARCH(1)	Normal	-8255.22	16516.45	16534.51	3

A.4.2 BIC ranking table

Table 15: Full BIC ranking of estimated volatility models.

Model	Distribution	Log-likelihood	AIC	BIC	# Parameters
EGARCH(1,1)	Student-t	-7609.20	15230.40	15266.52	6
EGARCH(1,1)	Skewed Student-t	-7609.13	15232.26	15274.40	7
EGARCH(1,1)	GED	-7627.92	15267.84	15303.96	6
GARCH(1,1)	Student-t	-7638.87	15287.74	15317.84	5
GARCH(1,1)	GED	-7639.46	15288.91	15319.01	5
GARCH(1,1)	Skewed Student-t	-7638.86	15289.72	15325.84	6
ARCH(3)	GED	-7747.68	15507.36	15543.48	6
ARCH(3)	Student-t	-7756.58	15525.16	15561.28	6
ARCH(3)	Skewed Student-t	-7756.58	15527.15	15569.29	7
ARCH(2)	GED	-7767.12	15544.24	15574.34	5
ARCH(2)	Student-t	-7778.16	15566.31	15596.42	5
ARCH(2)	Skewed Student-t	-7778.14	15568.29	15604.41	6
ARCH(1)	GED	-7795.42	15598.85	15622.93	4
ARCH(1)	Student-t	-7807.70	15623.40	15647.48	4
ARCH(1)	Skewed Student-t	-7807.69	15625.37	15655.48	5
EGARCH(1,1)	Normal	-8039.71	16089.43	16119.53	5
GARCH(1,1)	Normal	-8051.16	16110.32	16134.40	4
ARCH(3)	Normal	-8208.08	16426.16	16456.26	5
ARCH(2)	Normal	-8231.13	16470.26	16494.34	4
ARCH(1)	Normal	-8255.22	16516.45	16534.51	3

A.4.3 Residual Diagnostics

Table 16: Full Diagnostic results for estimated volatility models.

Model	Distribution	LB P-Value (standardized residuals)	LB P-Value (Squared Standardized Residuals)	LM P-Value
EGARCH(1,1)	Student-t	0.01	0.68	0.69
EGARCH(1,1)	Skewed Student-t	0.01	0.68	0.69
EGARCH(1,1)	GED	0.02	0.94	0.95
GARCH(1,1)	Student-t	0.01	0.91	0.91
GARCH(1,1)	GED	0.01	0.90	0.90
GARCH(1,1)	Skewed Student-t	0.01	0.91	0.91
ARCH(3)	GED	0.22	0.23	0.22
ARCH(3)	Student-t	0.22	0.37	0.35
ARCH(3)	Skewed Student-t	0.22	0.37	0.35
ARCH(2)	GED	0.20	< 0.01	< 0.01
ARCH(2)	Student-t	0.19	0.01	0.01
ARCH(2)	Skewed Student-t	0.19	0.01	0.01
ARCH(1)	GED	0.20	< 0.01	< 0.01
ARCH(1)	Student-t	0.19	< 0.01	< 0.01
ARCH(1)	Skewed Student-t	0.19	< 0.01	< 0.01
EGARCH(1,1)	Normal	0.03	0.94	0.94
GARCH(1,1)	Normal	0.04	0.85	0.85
ARCH(3)	Normal	0.25	0.02	0.02
ARCH(2)	Normal	0.27	< 0.01	< 0.01
ARCH(1)	Normal	0.22	< 0.01	< 0.01

A.4.4 Forecast Accuracy Table

Table 17: Full Forecast accuracy table for estimated volatility models.

Model	Distribution	RMSE	MAE	QLIKE
ARCH(1)	Normal	15.78	11.22	2.98
ARCH(1)	Student-t	30.12	26.05	3.55
ARCH(1)	GED	14.98	9.69	2.88
ARCH(1)	Skewed Student-t	30.09	26.01	3.55
ARCH(2)	Normal	15.60	10.68	2.94
ARCH(2)	Student-t	21.69	16.73	3.21
ARCH(2)	GED	15.07	9.19	2.83
ARCH(2)	Skewed Student-t	21.68	16.71	3.21
ARCH(3)	Normal	15.39	10.29	2.91
ARCH(3)	Student-t	18.78	13.52	3.05
ARCH(3)	GED	14.98	8.89	2.81
ARCH(3)	Skewed Student-t	18.78	13.52	3.05
GARCH(1,1)	Normal	14.77	8.31	2.75
GARCH(1,1)	Student-t	14.69	7.50	2.71
GARCH(1,1)	GED	14.63	7.39	2.70
GARCH(1,1)	Skewed Student-t	14.69	7.50	2.71
EGARCH(1,1)	Normal	14.71	8.73	2.77
EGARCH(1,1)	Student-t	16.72	11.10	2.90
EGARCH(1,1)	GED	14.64	7.66	2.72
EGARCH(1,1)	Skewed Student-t	16.74	11.12	2.90

A.4.5 Best models according to different criteria

Table 18: Model selected by each evaluation criterion.

Criterion	Selected model
AIC	EGARCH(1,1) — Student-t
BIC	EGARCH(1,1) — Student-t
RMSE	GARCH(1,1) — GED
MAE	GARCH(1,1) — GED
QLIKE	GARCH(1,1) — GED

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